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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCS101 - ADVANCED DATA STRUCTURES & ALGORITHMS

(For MCS)

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Module**.

Module – I

	Marks	BT*	CO*	PO*
1. a) Describe how a polynomial can be represented using (i) arrays and (ii) singly linked lists. For $P(x) = 5x^3 + 2x - 9$ show the storage structure in both cases and compare them.	10	3	1	2
b) Describe how a sparse matrix can be represented using a linked list of nodes. Write a C program to display all non-zero entries stored in the linked list.	6	3	1	2
c) With a neat sketch, explain the concept of a Circular Queue implemented using a dynamic array. Clearly state the condition that indicates that the queue is full.	4	2	1	2
2. a) Implement Stack and Queue using a Doubly Linked List (DLL). Write C functions to perform the following operations ensuring $O(1)$ insertion and deletion: (i) Stack: push() and pop() (ii) Queue: enqueue() and dequeue().	10	2	1	2
b) What is hashing? Explain with example. A hash table of size 11 uses: $h_1(k) = k \bmod 11$. Insert the keys 27, 33, 46, 57, 89, 90 using: 1. Linear probing 2. Quadratic probing For each method, write the probing sequence and the final table.	10	3	1	2

Module – II

3. a) Describe the structural differences between a B-Tree and a B+ Tree. Construct a B+ Tree of order 4 by inserting the keys 3, 7, 9, 12, 15, 20, 25, 30, 5, 10, showing all leaf and internal node splits, and explain how sequential access is supported.	10	3	1	2
b) What is a multiway search tree. Explain the search algorithm for a multiway search tree and apply it step-by-step on a tree to show how key comparisons determine the child pointer selected at each stage.	10	3	2	1
4. a) State the properties of AVL trees. Describe all four types of rotations used for balancing AVL trees with suitable diagrams, and analyze how rotations help maintain logarithmic height.	10	2	2	1
b) Write a C function to insert a node into a BST, construct a BST by inserting the keys 25, 17, 32, 13, 19, 28, 40, 14, 18 showing each step, and then delete only the nodes with two children one by one, displaying the tree after each deletion.	10	3	2	2

Module – III

5. a) Construct a recurrence tree for the relation $T(n)=T(n/4)+n$ and show how the tree is used to derive the overall asymptotic complexity of the recurrence relation.
- b) State and explain the five asymptotic notations used in algorithm analysis with suitable mathematical examples for each.

10 3 3 2

10 2 3 1

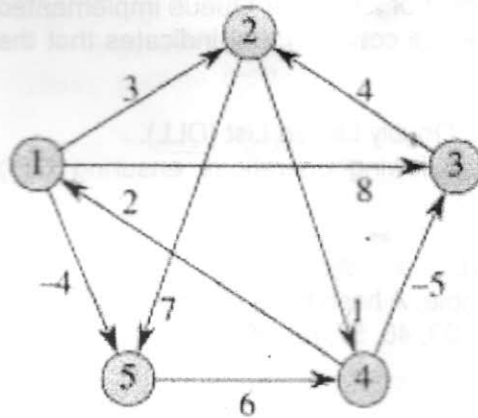
6. a) Use the Master Theorem to determine the time complexity of the recurrence $T(n)=4T(n/2)+n^2$ and explain each step of selecting the correct case of the theorem.
- b) State the Ratio Theorem and explain how it is used to compare the asymptotic growth rates of two functions $f(n)$ and $g(n)$. Using suitable examples, discuss all three possible outcomes of the limit and their implications on O , Ω , and Θ relationships.

10 3 3 2

10 2 3 1

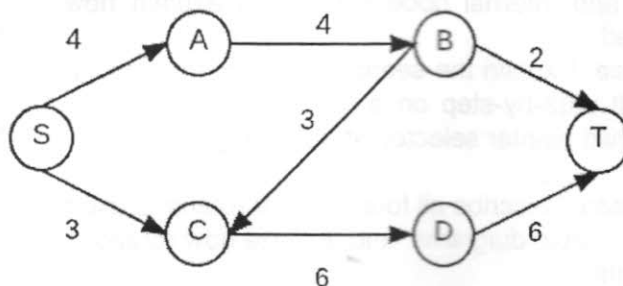
Module – IV

7. a) Outline the steps of Johnson's algorithm and explain how its reweighting mechanism enables shortest-path computation in sparse graphs. Then use this algorithm to determine the shortest-path distance between every pair of vertices for the graph given below.



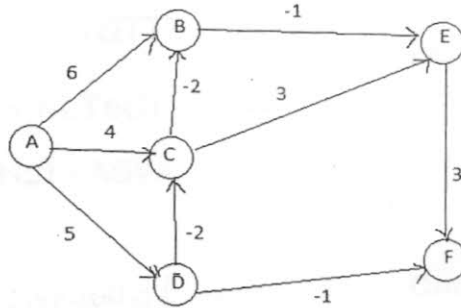
10 3 4 2

- b) Define a flow network and describe the key constraints that govern flows. Using these constraints, apply an appropriate maximum-flow algorithm to determine the maximum flow from the source to the sink for the flow network shown below.



10 3 4 2

8. a) Using the Bellman Ford algorithm find the shortest path from the source vertex to the remaining vertices in the graph shown below



10 3 4 2

- b) Explain how topological sorting helps in computing shortest paths in a DAG. Apply the algorithm to a DAG with at least 8 vertices and compute shortest paths from a given source.

10 3 4 2

Module – V

9. a) Using the Rabin–Karp algorithm with modulo $q = 11$ and radix $d = 10$, compute all hash values for the text “31415926535” and the pattern “26”. Identify spurious matches and actual matches.
- b) Compute the prefix function (π array) for the pattern “ABCDABD”. Using this π array, run the KMP algorithm on the text “ABCABCDABABCDABCDABDE” and show shifts and comparisons. Compare KMP and naïve matching in terms of pattern shifting, prefix reuse, and worst-case complexity.
- c) Explain randomizing a deterministic algorithm using Quick Sort as an example.
10. a) Define string matching problem. State a naïve string-matching algorithm and explain its working. Trace it for a given text acaabc and pattern aab.
- b) For the pattern “NEEDLE”, construct
(i) bad-character table
(ii) good-suffix table
Then apply Boyer–Moore on the text
“FINDINAHAYSTACKNEEDLEINASTACK”, showing all shifts.

5 2 5 1

10 3 5 2

5 2 5 1

10 2 10 1

10 3 5 2

BT* Bloom’s Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
